

Lecture 1

Transcendental Functions.

✓ We have mainly been using/dealing with functions called algebraic functions.

Examples

1. $f(x) = x^2 + 1$ ← polynomial

2. $f(x) = 1/x$ ← rational function

✓ These are built up by the usual operations of $+$, $-$, $\div$, $\neq$ and raising to constant.

✓ Functions that are not algebraic are called transcendental function.

✓ These cannot be expressed as solutions of poly. equations whose coefficients are themselves polys with rational coefficients.

✓ In fact, a transcendental number cannot be a root of a poly.

eg

$e, \sqrt{2}, \pi$ (proved until 1970's).

Examples of transcendental functions.

✓ Let c be a positive constant. The functions are transcendental.

- Exponential functions:

eg, $f(x) = x^\pi$

$g(x) = c^x$

$h(x) = x^x$

$k(x) = e^x$, etc

- Trigonometric functions:

eg, $\sin x, \cos x, \tan x, \csc x,$
 $\sec x, \arcsin x$, etc.

- Logarithmic functions:

eg $\log_c x, \ln x = \log_e x$

- Hyperbolic functions:

eg

$\cosh x, \sinh x, \tanh x, \operatorname{sech} x,$
etc.

Unfamiliar ones.

- Gamma

- * $\Gamma(n) = (n-1)! \quad \forall \quad n \text{ positive integer}$

- * $\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt$

hypertranscendental
functions

- Zeta fn

- * $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$

$$= 1 + \frac{1}{2^s} + \frac{1}{3^s} + \dots$$

Logarithmic and exponential functions e^x $\exp(x)$

✓ The "real" exponential function

$\exp: \mathbb{R} \rightarrow \mathbb{R}$
can be characterized in several ways.

1. Using / as a power series

$$e^x = \exp(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!} \quad \text{Later!}$$

$$= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \leftarrow$$

✓ We henceforth get it that $e^0 = 1$

$$\frac{d}{dx}(f+g) = \frac{df}{dx} + \frac{dg}{dx}$$

$$\begin{aligned} e &= \exp(1) \\ &= \sum_{k=0}^{\infty} \frac{1}{k!} \end{aligned}$$

✓ A term by term differentiation of $\exp(x)$ reveals that

$$\boxed{\frac{d(e^x)}{dx} = e^x} \quad \forall x \in \mathbb{R}$$

Why and how?

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120} + \dots$$

$$\begin{aligned} \frac{d(e^x)}{dx} &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \\ &= e^x \end{aligned}$$

2. As a limiting value

$$e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n$$

Proof!

Later!

Lecture 2

1^∞

a^x

3. Via the natural log function

Properties of exp function.

- The usual ones

Proofs (Later!)

eg

$$* e^x \cdot e^y = e^{x+y}$$

$$* (e^x)^y = e^{xy}$$

$$* \frac{e^x}{e^y} = e^{x-y}$$

$$* e^{-x} = 1/e^x$$

The natural log function.

Recall

$$y = \log_a x \iff x = a^y \quad *$$

- The following properties hold

$$* \log_a pq = \log_a p + \log_a q$$

$$* \log_a \left(\frac{p}{q} \right) = \log_a p - \log_a q$$

$$* \log_a p^r = r \log_a p \quad \text{where } p, q \text{ are the}$$

real numbers and r is any real number

✓ Recall

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad **$$

✓ What is $\int x^{-1} dx$?

$$\ln x = \log_e x$$

Recall

$$\frac{d}{dx} (\log_a x) = \frac{1}{x \ln a} \quad *$$

$$\int \frac{1}{x \ln a} dx = \log_a x + C$$

Definition

For all $x > 0$, we define

$$* \int_1^x \frac{1}{t} dt = \ln x$$

$$\boxed{\text{F.T.C}}$$

- ✓ The expression $\ln x$ is called the natural logarithm of x .
- ✓ x has to be positive since $\int_1^x \frac{1}{t} dx$ does not exist when $x \leq 0$.

✓ The F.T.C implies that

$$\frac{d}{dx} \left(\int_1^x \frac{1}{t} dt \right) = \frac{d}{dx} (\ln x)$$

$$\boxed{\frac{1}{x} = \frac{d}{dx} (\ln x)} **$$

✓ We can also deduce that

$$\ln 1 = \int_1^1 \frac{1}{t} dt = 0$$

$$\boxed{\ln 1 = 0}$$

✓ ** agrees with the following.

$$* \ln e^x = \log_e e^x$$

$$* \boxed{\ln e^x = x} \quad \text{with } \log_e e \text{ circled and an arrow pointing to it from } \ln e$$

$$\boxed{\ln e = 1}$$

$$* \boxed{\log_a x = \frac{\ln x}{\ln a}}$$

- We said that $y = \log_a x \Leftrightarrow \boxed{x = a^y}^{**}$

$$\Leftrightarrow \ln x = \ln a^y$$
$$\ln x = y \ln a^{**}$$

$$\Leftrightarrow \textcircled{y} = \frac{\ln x}{\ln a}$$

$$\Rightarrow \log_a x = \frac{\ln x}{\ln a}$$

$$\frac{d}{dx} (\log_a x) = \frac{d}{dx} \left(\frac{\ln x}{\ln a} \right)$$

$$= \frac{1}{\ln a} \frac{d}{dx} (\ln x)$$

$$= \frac{1}{\ln a} \cdot \frac{1}{x}$$

$$\frac{d}{dx} (\log_a x) = \frac{1}{x \ln a}$$

Let $a = e$.

So that

$$\frac{d}{dx} (\log_e x) = \frac{d}{dx} (\ln x) = \frac{1}{x \ln e}$$

$$\boxed{\frac{d}{dx} (\ln x) = \frac{1}{x}}^{**}$$

Defn

The natural exponential fn, denoted by $\exp(x)$ is defined by

$$y = e^x \Leftrightarrow \ln y = x$$

for all x , where $y > 0$.

✓ We notice that $\ln$ and $\exp$ are inverses of each other.
why?

$$\begin{aligned}\ln e^x &= x \quad \forall x \\ e^{\ln y} &= y \quad \text{for all } y > 0\end{aligned}$$

$$f(g(x)) = x$$

$$g(f(x)) = x$$

$\Updownarrow$
 f and g are
inverses of each
other.